

Document reference		NPS/001/017						
Version:-	2.0	Date of Issue:-	May 2009	Page	1	of	14	

NPS/001/017 – Technical Specification for Low Voltage and High Voltage Current Limiting Fuse Links

1. Purpose

The purpose of this document is to detail the technical requirements for high voltage (HV) and low voltage (LV) fuses for use on the distribution networks of CE Electric UK.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/011/007	1.0	Specification for HV and LV cartridge Fuse Links
TS E013/001	2001	HV and LV HRC and expulsion Current Limiting Fuse Links
NPS/001/017	1.0	Technical specification for LV and HV current limiting fuse links

2. Scope

This document applies to all fuses for use on the distribution networks of CE Electric UK. These fuses shall be manufactured in accordance with the appropriate national and international standards as detailed within the specification. The following appendixes have been included within this specification: -

- Appendix 1 – Schedule of Items required by CE Electric UK.
- Appendix 2 – Self certification conformance declaration that is to be completed by the supplier. This document will provide details of conformance or otherwise with the appropriate national or international standard.

3. Technical Requirements

3.1 Range of Products

The range of products covered in this specification is included in the following sub-sections with specific dimensional requirements in Appendix 1.

3.1.1 High Voltage Current Limiting Fuses

HV current limiting fuses shall comply with the technical requirements of BS EN 60282-1: 2006 and application requirements of ENATS 12-8 Issue 2: 1986. They are required for operation under oil and for operation in air in normal service conditions as detailed in BS EN 60282 – 1 Section 2.1.

Document reference		NPS/001/017						
Version:-	2.0	Date of Issue:-	May 2009	Page	2	of	14	

Fuse links are required for 11kV and 20kV systems with the rated voltage in accordance with Series 1 of BS EN 60282 – 1 Table 3 and a system frequency of 50Hz.

HV fuse links for use under oil shall conform to the dimensional requirements of BS EN 60282 – 1, Annex D Type II, dimension A = 63.5mm and two different lengths of dimension D = 256mm and also D = 361mm.

HV fuse links for use in air shall conform to the dimensional requirements of BS EN 60282 – 1, Annex D Type III, dimension A = 80mm and E = 419mm with Fig. D type fixing holes.

Fuse links rated at 24kV shall be oil-immersed type with a length of 361mm and shall have a breaking capacity of not less than 12.5kA.

All HV fuse links shall be fitted with strikers for operation of the switch fuse tripping mechanism. The striker operation shall be designed to comply with the requirements of BS EN 62271 – 105: 2004. The direction of striker pin operation shall be clearly marked on the fuse link.

3.1.2 High Voltage Expulsion Fuse Links

HV expulsion fuse links shall be suitable for use in expulsion fuse switch equipment as detailed in ENA TS 41-36 Issue 2 - 2004.

Fuse links are required for use on 11kV, 20kV and 33kV distribution networks and shall have a rated current breaking capacity of 8kA.

The links shall comply with BS 2692 – 2: 1956. The following variations of fuse element are required: -

- Slow blow with tails at both ends.
- Slow blow universal with removable NEMA button and tail on one end with tail at the other.
- Slow blow threaded with ¼ inch UNF screw thread (to allow the use of an extension rod) at one end with a removable NEMA button and a tail at the other.

ANSI C 37-2 Type T slow blowing characteristics are required with the overall fuse lengths are detailed in Appendix 1.

3.1.3 High Voltage DIN Type Full Range Fuse Links

HV DIN type fuse links are for use in padmount type transformers with no associated HV switchgear. Full Range type fuse links are required in accordance with BS EN 60282 – 1, Section 3.3.5. Dimensions shall be in accordance with DIN 43 625 and are detailed in Appendix 1.

3.1.4 Low Voltage Current-limiting Fuse-links.

LV current-limiting “J” type fuse-links shall comply with the performance and testing requirements of BS EN 60269 – 1: 2007. They shall be full range breaking-capacity with the rated current prefixed with the letter “g” as detailed in BS EN 60269 – 1 Section 5.7.1. Dimensions shall be to BS 88 – 2: 2007, Figure 905, with L type tags.

3.1.5 Low Voltage Domestic Fuse Links

Fuse-links for domestic and similar premises shall comply with the requirements of BS 88 – 3:2007. CE Electric UK has a requirement for Types IIa and IIb as detailed in Figure 301 of BS 88 - 3.

Document reference		NPS/001/017						
Version:-	2.0	Date of Issue:-	May 2009	Page	3	of	14	

Fuse-links shall be suitable for use in cut-outs designed to typical design requirement of BS 88 – 3, Figure 303.

3.1.6 Low Voltage Street Lighting Fuse-links

Fuse-links for street lighting cut-out applications shall comply with the general requirements of BS EN 60269 – 1: 2007 and specific requirements of BS 7654: 1997. The fuse shall have an offset tag with 38mm fixing centres and have a minimum breaking capacity of 16kA in accordance with Fig. 102 of BS 7654.

3.1.7 Low Voltage Industrial and Commercial Fuses

Industrial and commercial cartridge fuses shall be in accordance with BS 88 – 2 and BS EN 60269 – 1. Appendix 1 details the ratings and BS 88 – 2 references.

3.2 Markings

HV fuse links shall have identification markings in accordance with BS EN 60282 – 1, Section 5.2. LV fuse links shall have identification markings in accordance with BS EN 60269 – 1, Section 6.2.

3.3 Testing

Testing shall be in accordance with the appropriate IEC or British Standard. Suppliers shall complete the self certification conformance declaration detailed in Appendix 2.

Document reference		NPS/001/017					
Version:-	2.0	Date of Issue:-	May 2009	Page	4	of	14

4. References

4.1 External Documentation

Reference	Title
BS EN 60282: 1 - 2006 ENATS 12 – 8	High-voltage fuses. Current-limiting fuses The Application of Fuse Links to 11kV/425v Underground Distribution Networks
BS EN 62271:105 – 2003	High-voltage switchgear and control gear. Alternating current switch-fuse combinations
BS EN 60269: 1 – 2007 BS 88: 1 - 2007 BS 88:3 - 2007	Low-voltage fuses. General requirements Supplementary requirements for fuses for use by unskilled persons (fuses mainly for household or similar applications)
BS 88: 2 - 2007	Supplementary requirements for fuses for use by authorised persons (mainly for industrial applications)
BS 7654: 1 - 1997	Specification for single phase street lighting fuses (cut- outs) for low-voltage public electricity distribution systems. 25 A rating for highway power supplies and street furniture
BS 2692: 2 - 1956	Fuses for voltages exceeding 1000 V a.c. - Expulsion fuses

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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Document reference		NPS/001/017						
Version:-	2.0	Date of Issue:-	May 2009	Page	5	of	14	

6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards Engineer	S Salkeld	20/04/2009

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	06/05/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead System Standards Manager	G Hammel	06/05/2009

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	26/05/2009

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessments Manager	C Holdsworth	06/05/2009

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	20/05/2009

Document reference	NPS/001/017				
Version:-	2.0	Date of Issue:-	May 2009	Page	6 of 14

Appendix 1 – Schedule of Items

HV Current Limiting Fuses in Oil (Section 3.1.1)

System Rated Voltage (kV)	Current Rating (A)	ENA TS 12-8 rating Ref.	Length -Dimension D (mm)	Commodity Code
11	80	O4	359	287788
11	90	O5	359	287805
11	80	O4	254	038620
11	31.5	O2	254	287699
11	40	O3	254	083604
11	63	O4	254	038612
11	50	O3	254	287716
11	20	O1	254	288009
20	10	N/A	361	280011
20	20	N/A	361	287824
20	31.5	N/A	361	287839
20	50	N/A	361	287843

HV Current Limiting Fuses in Air (Section 3.1.1)

System Rated Voltage (kV)	Current Rating (A)	ENA TS 12-8 rating Ref.	Length -Dimension D (mm)	Commodity Code
11	20	A1	359	287595
11	31.5	A2	359	287608
11	50	A3	359	287612
11	71	A4	359	287627

HV Expulsion Fuse Links (Section 3.1.2)

Current Rating (A)	Fuse Type	System Voltage (kV)	Length (mm)	Commodity Code
100	Slow Blow	11	550	037879
100	Slow Blow	33	800	037895
6	Slow Blow, Universal	11, 20, 33	787	245840
10	Slow Blow, Universal	11, 20, 33	787	245037
12	Slow Blow, Universal	11, 20, 33	787	245041
15	Slow Blow, Universal	11, 20, 33	787	245056
25	Slow Blow, Universal	11, 20, 33	787	245018
30	Slow Blow, Universal	11, 20, 33	787	242518
40	Slow Blow, Universal	11, 20, 33	787	245060
65	Slow Blow, Universal	11, 20, 33	787	245446
6	Slow Blow, Threaded	11, 20	533	259006
10	Slow Blow, Threaded	11, 20	533	259010
12	Slow Blow, Threaded	11, 20	533	259025
15	Slow Blow, Threaded	11, 20	533	259044
25	Slow Blow, Threaded	11, 20	533	259059
30	Slow Blow, Threaded	11, 20	533	259063
40	Slow Blow, Threaded	11, 20	533	259078

Document reference	NPS/001/017						
Version:-	2.0	Date of Issue:-	May 2009	Page	7	of	14

Current Rating (A)	Fuse Type	System Voltage (kV)	Length (mm)	Commodity Code
50	Slow Blow, Threaded	11, 20	533	259082
65	Slow Blow, Threaded	11, 20	533	259097
80	Slow Blow, Threaded	11, 20	533	259114
100	Slow Blow, Threaded	11, 20	533	259129

HV DIN Type Fuse Links (Section 3.1.3)

System Voltage (kV)	Current rating (A)	DIN 43 625 Length (mm)	Commodity Code
11	6.3	292	362207
11	10	292	362194
11	20	292	362175

LV Current Limiting Fuse-links – “J” Type Cylindrical (Section 3.1.4)

Current Rating (A)	Dimensions	Commodity Code
100	Length 45.7mm – Diameter 38.7mm (no tag)	031625
150	Length 45.7mm – Diameter 38.7mm (no tag)	031633
200	Length 45.7mm – Diameter 38.7mm (no tag)	031641
200	76mm - Dimension A - BS 88-2 Fig. 905	280562
63	76mm - Dimension A - BS 88-2 Fig. 905	280702
100	82mm - Dimension A - BS 88-2 Fig. 905	280721
160	82mm - Dimension A - BS 88-2 Fig. 905	280740
200	82mm - Dimension A - BS 88-2 Fig. 905	280774
250	82mm - Dimension A - BS 88-2 Fig. 905	280628
315	82mm - Dimension A - BS 88-2 Fig. 905	280632
400	82mm - Dimension A - BS 88-2 Fig. 905	280666
500	82mm - Dimension A - BS 88-2 Fig. 905	035568
100	92mm - Dimension A - BS 88-2 Fig. 905	280789
160	92mm - Dimension A - BS 88-2 Fig. 905	280793
200	92mm - Dimension A - BS 88-2 Fig. 905	280825
250	92mm - Dimension A - BS 88-2 Fig. 905	280384
315	92mm - Dimension A - BS 88-2 Fig. 905	280844
355	92mm - Dimension A - BS 88-2 Fig. 905	280399
Current Rating (A)	Dimensions	Commodity Code
400	92mm - Dimension A - BS 88-2 Fig. 905	280863
500	92mm - Dimension A - BS 88-2 Fig. 905	280878
630	92mm - Dimension A - BS 88-2 Fig. 905	280952

Domestic fuse Links (Section 3.1.5)

Current Rating (A)	End Cap Diameter – BS 88 - 3 Table 301 (mm)	Type – BS 88 - 3 Table 301	Commodity Code
5	22.23	IIa	282318
25	22.23	IIa	181051
30	22.23	IIa	031088
40	22.23	IIa	282360
60	22.23	IIa	282394

Document reference		NPS/001/017					
Version:-	2.0	Date of Issue:-	May 2009	Page	8	of	14

Current Rating (A)	End Cap Diameter – BS 88 - 3 Table 301 (mm)	Type – BS 88 - 3 Table 301	Commodity Code
80	22.23	Ila	282407
30	30.16	Ilb	031203
60	30.16	Ilb	282430
80	30.16	Ilb	282426
100	30.16	Ilb	282411

Street Lighting Fuse Links (Section 3.1.6)

Current rating (A)	Dimension G – BS 7654 Fig. 102 (mm)	Tag Description	Commodity Code
6	38	Offset tags with 1 axial and 1 lateral slot	282750
10	38	Offset tags with 1 axial and 1 lateral slot	282765
16	38	Offset tags with 1 axial and 1 lateral slot	282784
20	38	Offset tags with 1 axial and 1 lateral slot	282799
25	38	Offset tags with 1 axial and 1 lateral slot	030718
32	38	Offset tags with 1 axial and 1 lateral slot	030726

LV Industrial Fuse Links (Section 3.1.7)

System Voltage (V)	Current Rating (A)	Tag Arrangement	Commodity code
440	400	BS 88-2 Special Offset 92.5mm Centres With Open Double Slot	032813
440	630	BS 88-2 Special Offset 94.0mm Centres With Open Double Slot	032870
415	6	BS 88-2 Ref. F1	032078
415	10	BS 88-2 Ref. F1	032094
415	16	BS 88-2 Ref. F1	032102
415	32	BS 88-2 Ref A2	032508
415	63	BS 88-2 Ref A2	032573
415	400	BS 88-2 Ref C1	034157
415	400	BS 88-2 Ref B4	032902
145	500	BS 88-2 Ref C2	034322

Chemical Actuator (for Cooper Bussmann Automatic Sectionalisers Link)

System Voltage (kV)	Supplier Reference	Commodity Code
11	E2906	102988

Document reference		NPS/001/017					
Version:-	2.0	Date of Issue:-	May 2009	Page	9	of	14

Appendix 2

SELF CERTIFICATION CONFORMANCE DECLARATION

LV and HV current limiting fuse links are covered by IEC and British Standards and shall comply with the latest issues.

This check sheet identifies the clauses of the aforementioned Standards relevant to LV and HV current limiting fuse links for use on the CE Electric UK distribution network.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or 'ENATS' as appropriate

Manufacturer:

Product Reference:

Name:

Signature:

Date:

Document reference		NPS/001/017					
Version:-	2.0	Date of Issue:-	May 2009	Page	10	of	14

Schedule 2 Table

HV Current Limiting Fuses (Section 3.1.1)

• BS EN 60282 - 1			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
2.0	Service Conditions		
4.0	Ratings		
5.0	Design Construction and Performance		
7.0	Type Tests		
Annex D	Dimensions		
• BS EN 62271 - 105			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
Various	Direction of Striker Pin		

LV Current Limiting Fuse Links (Section 3.1.2)

• BS EN 60269 - 1			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
4.0	Classification		
5.7	Breaking Range and Capacity		
6.2	Marking of Fuse Links		
8.0	Testing		
• BS 88 - 5			
5.9	Dimensions		

Document reference		NPS/001/017					
Version:-	2.0	Date of Issue:-	May 2009	Page	11	of	14

Domestic Fuse Links (Section 3.1.3)

• BS 88 - 3			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
4	Classification		
5	Characteristics of fuses		
6	Markings		
7	Standard conditions for construction		
8	Tests		

Street Lighting Fuse Links (Section 3.1.4)

• BS EN 60269 - 1			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
4.0	Classification		
5.7	Breaking Range and Capacity		
6.2	Marking of Fuse Links		
8.0	Testing		
• BS 7654			
5.9	Figure 102 – Dimensional Requirements		

Document reference		NPS/001/017					
Version:-	2.0	Date of Issue:-	May 2009	Page	12	of	14

HV Expulsion Fuse Links (Section 3.1.5)

- **BS 2692 - 2**

Clause/Sub-clause	Requirement	Conformance Code	Remarks
Section 2 - 4,5,6	Ratings		
Section 3 – 7	Markings		
Section 6 – 17	Construction		
Section 7	Type Tests		
Section 8	Routine Tests		

HC DIN Type Full range Fuse Links (Section 3.1.6)

- **BS EN 60282 - 1**

Clause/Sub-clause	Requirement	Conformance Code	Remarks
2.0	Service Conditions		
4.0	Ratings		
5.0	Design Construction and Performance		
7.0	Type Tests		

- **DIN 43 625**

Clause/Sub-clause	Requirement	Conformance Code	Remarks
	Dimensional Requirements		

Document reference		NPS/001/017					
Version:-	2.0	Date of Issue:-	May 2009	Page	13	of	14

Industrial and Commercial Cartridge Fuses (Section 3.1.7)

• BS EN 60269 - 1			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
4.0	Classification		
5.7	Breaking Range and Capacity		
6.2	Marking of Fuse Links		
8.0	Testing		
• BS 88 - 2			
Figure 1	Dimensions for Reference A, B, C and D Types		

Document reference		NPS/001/017						
Version:-	2.0	Date of Issue:-	May 2009	Page	14	of	14	

Appendix 3 - Proof of Performance

Evidence of proof of performance shall be submitted for each type of fuse link offered. This shall include Time/Current characteristics and curves in compliance with the relevant IEC and British Standards. A sample of each product group offered shall be provided for assessment.